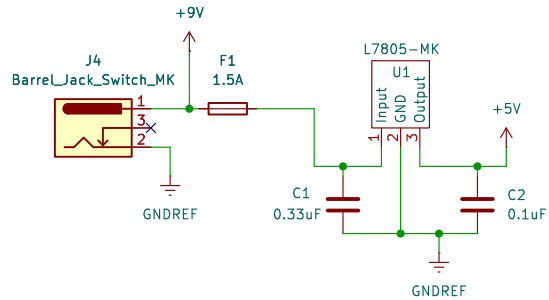


All Resistors rated for 1/4 Watt
All Capacitors rated for 50V

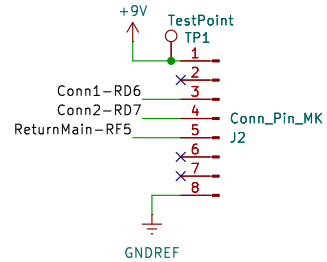
Power supply is not needed, however is here for testing purposes, but the main power will be jumped from Kelton (main board).

Power Supply Circuit



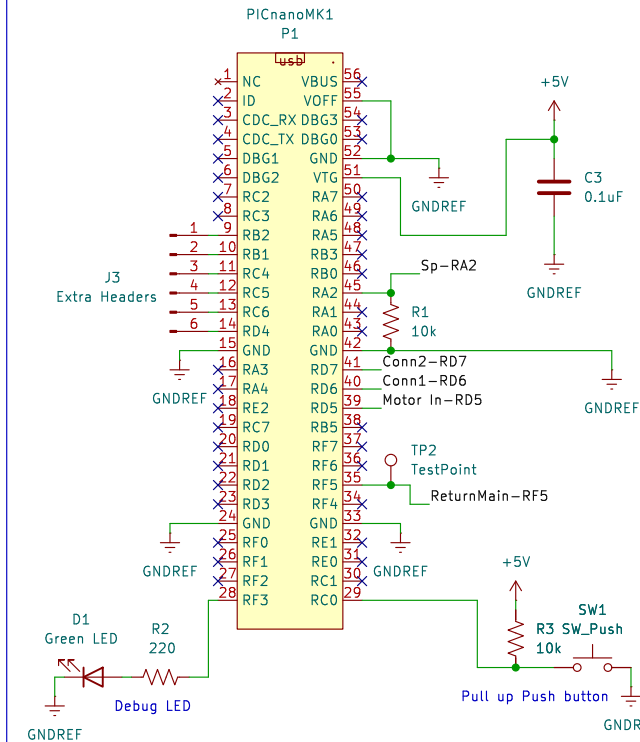
In connector circuit, two PWM pins RD6 and RD7 connect to Microcontroller and RF5 is the pin signal going back to the Main Board (Kelton). The group will run a shared power supply of 9v, also we have common ground.

Subsystem Connector Circuit



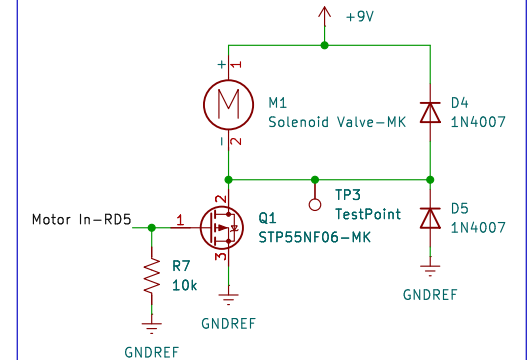
Microchip microcontroller receives signals from keltons board and reroutes them to the solenoid and speaker. RF5 is going to be a return signal to main board (Kelton), this will not actually do anything it's for debugging process or extra signal in case of emergency. For other debugs I have the led and push button. Also there are extra headers in case of failures.

MicroController Circuit



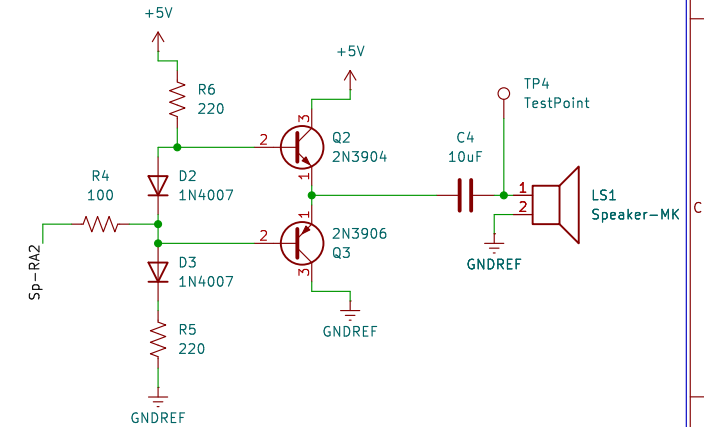
The solenoid receives the signal from RD5 and will be converted from microchips RD6 pin. So RD6 is received from Keltons board and converted to RD5 to open the solenoid. The diodes are a freewheeling diode. A MOSFET is also incorporated for the solenoid.

Solenoid Valve Circuit



The speaker receives an analog input from the microchip to play multiple sounds. The microchip takes a signal from RD7 and reroutes it into analog from RA0 and pushes through the speaker circuit to play sound.

Speaker Circuit



Team 211 - EGR 304

Sheet: /

File: SubsystemMK.kicad_sch

Title: MK Subsystem

Size: A4 Date: 2025-11-01

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